

ZERÔPÂ■ MONEY — TECHNICAL CREDIBILITY ABSTRACT

Redacted 2-Page Summary — for lab validation partners, acquirers, and technical due diligence, pre-NDA

DRAFT — Aug 13, 2026 — awaiting Tom's review before any send. Proves depth of understanding without disclosing the method.

1. The Problem — Why Contactless Payments Are Capped

Every contactless payment scheme in the world today enforces a low per-transaction value ceiling for tap-and-go without a PIN (in the UK, currently £100; other markets set their own equivalent). This cap exists because the underlying Consumer Device Cardholder Verification Method (CDCVM) model was never designed to guarantee security for high-value transactions without a live, real-time connection back to the issuer to check the transaction. Take that live connection away — as is the case across most of rural Nigeria and much of the underserved world — and the cap becomes even more conservative, because none of the usual safety nets (velocity checks, real-time fraud scoring, issuer decline) are available at all.

ZerôPâ■ Money was built to solve this specific problem: how do you make a contactless tap safe, at any value, with zero live connectivity — not just intermittent connectivity, but genuine multi-day offline operation in dead zones.

2. Origin and Credentials

This is not a new idea chasing a trend. The foundational thinking traces back to a 2011 MSc dissertation in Computer Systems Security, University of Gloucestershire — NFC Contactless Technology for Financial Payments — completed with Merit distinction. What has changed since 2011 is what hardware now makes possible: physically unclonable function (PUF) silicon and modern elliptic-curve cryptography have matured from academic concepts into commercially deployable, low-cost components. The current build is best understood as an evolution of that original thinking, engineered against hardware that did not exist in a commercially viable form at the time of the original dissertation.

3. The Six Attack Vectors — Addressed

Any credible offline high-value contactless system has to answer six specific attack categories. ZerôPâ■ Money's position, stated at the level any lab or acquirer needs to evaluate it — without disclosing the underlying mechanism:

1. Eavesdropping — intercepted signal data yields nothing usable to an attacker, even with dedicated radio-analysis hardware (SDR, Proxmark-class tools).
2. Relay attack — transactions carried out at distance fail; the protocol enforces a tight physical proximity check.
3. Man-in-the-middle — both sides of a transaction cryptographically confirm each other before any payment data is exposed.
4. Replay attack — every transaction is single-use and time-bound; a captured transaction cannot be resubmitted.

5. Offline double-spend — the system guarantees against double-spending with zero live network validation, the core unsolved problem in offline EMV since at least 2011.

6. CDCVM bypass — the cardholder verification step cannot be circumvented to force an unauthorized high-value tap through.

This is the exact claim a formal attack-test report (FIME, UL, or equivalent) would be commissioned to independently verify: that all six hold under real adversarial testing — not how they are achieved.

4. IP Status

ZerôPâ■ Money's method is patent-pending. Full technical disclosure of the mechanism is reserved for formal, NDA-protected engagement — consistent with how any early-stage hardware-security IP is handled ahead of a completed filing and lab-validated attack report.

5. What a High-Value Tap Actually Looks Like

For the customer: identical to any contactless tap today — hold phone or card near the terminal, done in under two seconds, no PIN entry, no network wait. For the merchant: instant confirmation on the terminal display, with the transaction cryptographically finalized and later synced to settlement rails the next time any device in the network reaches connectivity — whether that is minutes or several days later. The value of the transaction is not artificially limited by a global scheme cap; it is limited only by the merchant's and issuer's own risk policy, the same way a \$1,000,000 chip-and-PIN transaction is possible today with a PIN, just without needing one.

This is the same evolutionary path Apple Pay itself took to raise contactless limits in some markets: one acquirer, one approved high-value MCC under a proven kernel, before any global scheme rule changes — not a request to rewrite Visa or Mastercard's global rulebook on day one.

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